

W1

Threat Detection Platform

5
DAYS

Project Information

Project	Threat Detection Platform
Organization	Infotact Solutions & Co.
Domain	Cybersecurity — Threat Detection Platform
Week	Week 1 of 4
Stack	Python + MongoDB + VirusTotal + GitHub
Duration	5 Days (Day 1 to Day 5)
Version	v1.0.0

Week 1 Objectives

- fetch threat intelligence from VirusTotal
- extract malicious IP reputation data
- store structured threat records in MongoDB

DAY 1

System Setup + Core Tools Installation

STEP 1 — Install Python 3 (Goal: Python runtime installation)

<https://www.python.org/downloads/>
Enable during installation: Add Python to PATH, Install pip, Install for all users
python --version
pip --version

Expected Output: Python 3.13.x / pip 24.x.x

STEP 2 — Install Visual Studio Code (Goal: Code editor setup)

<https://code.visualstudio.com/>
Install extensions: Python, Pylance, MongoDB for VS Code, GitLens

Expected Output: VS Code opens with integrated terminal

STEP 3 — Install Git (Goal: Version control setup)

<https://git-scm.com/downloads>
git --version

Expected Output: git version 2.x.x

DAY 2 Project Structure + Python Environment

STEP 4 — Create Project Folder (Goal: Project directory structure)

```
cd Desktop
mkdir Threat_detection
cd Threat_detection
dir
```

Expected Output: Threat detection

STEP 5 — Create Python Virtual Environment (Goal: Isolated Python environment)

```
python -m venv venv
venv\Scripts\activate
where python
```

Expected Output: ...\\Threat detection\\venv\\Scripts\\python.exe

STEP 6 — Install Required Packages (Goal: Project dependencies)

```
pip install requests pymongo python-dotenv
pip freeze > requirements.txt
```

Expected Output: Successfully installed requests pymongo python-dotenv

DAY 3 MongoDB Setup + Database Connection

STEP 7 — Install MongoDB Community Server (Goal: Local database server)

```
https://www.mongodb.com/try/download/community
Enable during installation: Complete Setup, Install MongoDB as a Service, Install MongoDB Compass
services.msc
Find: MongoDB
```

Expected Output: MongoDB service visible

STEP 8 — Start MongoDB Service (Goal: Start local MongoDB server)

```
net start MongoDB
mongodb://localhost:27017
```

Expected Output: The MongoDB service was started successfully.

STEP 9 — Create Database Connection File (Goal: MongoDB connection setup)

```
from pymongo import MongoClient
client = MongoClient("mongodb://localhost:27017/")
db = client["threat_intel"]
collection = db["threats"]
```

Expected Output: MongoDB connection ready

DAY 4

VirusTotal API + Threat Collection

STEP 10 — Configure Environment Variables (Goal: Secure API configuration)

```
VT_API_KEY=your_api_key_here
import os
from dotenv import load_dotenv
load_dotenv()
VT_API_KEY = os.getenv('VT_API_KEY')
```

Expected Output: API key loaded securely

STEP 11 — Create Threat Fetch Script (Goal: Fetch threat intelligence)

```
import requests
from config import VT_API_KEY
from db import collection
from datetime import datetime

url = "https://www.virustotal.com/api/v3/ip_addresses/8.8.8.8"
headers = {"x-apikey": VT_API_KEY}

response = requests.get(url, headers=headers)
data = response.json()

result = {
    "ip": "8.8.8.8",
    "malicious_score": data["data"]["attributes"]["last_analysis_stats"]["malicious"],
    "timestamp": datetime.utcnow()
}

collection.insert_one(result)
print(result)
```

Expected Output: {'ip': '8.8.8.8', 'malicious_score': 0}

DAY 5

Run Project + Verify Output

STEP 12 — Run Threat Fetching + Verify MongoDB (Goal: Validate Week 1 pipeline)

```
python fetch_threats.py
mongodb://localhost:27017
Open in MongoDB Compass: threat_intel → threats
```

Expected Output: {"ip":"8.8.8.8","malicious_score":0,"timestamp":"2026-..."}

Completion Checklist

- ✓ Python installed and working
- ✓ VS Code installed and configured
- ✓ Git installed and working
- ✓ Project folder created
- ✓ Virtual environment created and activated
- ✓ Required packages installed
- ✓ MongoDB installed and running
- ✓ VirusTotal API configured
- ✓ Threat data fetched successfully
- ✓ MongoDB contains threat records

Week 1 — Required GitHub Commits

Commit Message

<code>git commit -m "feat: init project structure and environment setup"</code>
<code>git commit -m "feat: add MongoDB database connection"</code>
<code>git commit -m "feat: add VirusTotal API configuration"</code>
<code>git commit -m "feat: add threat ingestion script"</code>
<code>git commit -m "test: verify threat ingestion pipeline"</code>